

Smart Parking System Using ESP32

Objective

The objective of this project is to detect whether a parking slot is available or occupied using an ultrasonic sensor and indicate the status using LEDs.

Components Used

- ESP32 DevKit V1
- HC-SR04 Ultrasonic Sensor
- Green LED
- Red LED
- $2 \times 220\Omega$ Resistors
- Jumper Wires

Sensors Used

HC-SR04 Ultrasonic Sensor – It measures the distance between the sensor and an object (vehicle) using ultrasonic waves.

Working Principle

The HC-SR04 continuously measures the distance in front of it. If a vehicle is detected within 10 cm, the ESP32 turns on the red LED to indicate that the parking slot is occupied. If the distance is greater than or equal to 10 cm, the ESP32 turns on the green LED to indicate that the parking slot is available.

Circuit Connections

- HC-SR04 VCC → ESP32 VIN (5V)
- HC-SR04 GND → ESP32 GND
- HC-SR04 TRIG → GPIO 5
- HC-SR04 ECHO → GPIO 18
- Green LED (+) → 220Ω Resistor → GPIO 2
- Green LED (–) → GND

- Red LED (+) → 220Ω Resistor → GPIO 4
- Red LED (-) → GND

Applications

- Shopping Mall Parking
- Hospital Parking
- Office Parking
- College and University Parking
- Smart City Parking Systems

Advantages

- Easy to build
- Low cost
- Accurate vehicle detection
- Reduces parking time
- Can be upgraded for IoT applications

Future Scope

- Add Wi-Fi monitoring using ESP32
- Display parking status on a mobile app
- Count total available parking slots
- Send notifications when a slot becomes available

Result

The Smart Parking System was successfully implemented using ESP32. The ultrasonic sensor detected the presence of a vehicle, and the LEDs correctly indicated whether the parking slot was available or occupied.

Project Link

<https://wokwi.com/projects/469993687320184833>